

1.6 The Cross Product

$$\mathbf{A} \times \mathbf{B} = a_N |\mathbf{A}| |\mathbf{B}| \sin \theta_{AB}$$

a_N is normal unit vector

Important properties of cross product or vector product:

a) Screw rule: The direction will be given by screw rule as shown in Fig. 1.7.

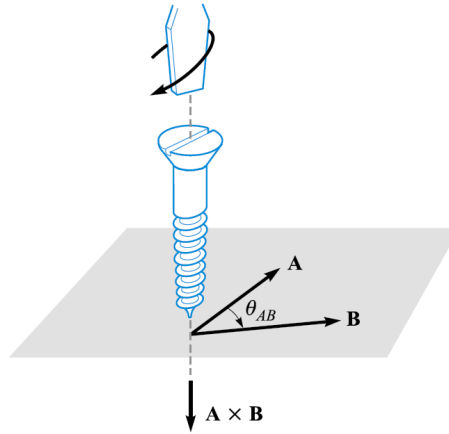


Fig. 1.7 The direction of $\mathbf{A} \times \mathbf{B}$ is in the direction of advance of a right-handed screw as $\mathbf{A}$ is turned into $\mathbf{B}$.

b) $\mathbf{F} = I(\mathbf{L} \times \mathbf{B})$

c) $\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \mathbf{a}_x + (A_z B_x - A_x B_z) \mathbf{a}_y + (A_x B_y - A_y B_x) \mathbf{a}_z$ or written as a determinant in a more easily remembered form,

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Thus, if $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = -4\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$, we have

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 2 & -3 & 1 \\ -4 & -2 & 5 \end{vmatrix} \\ &= [(-3)(5) - (1)(-2)]\mathbf{a}_x - [(2)(5) - (1)(-4)]\mathbf{a}_y + [(2)(-2) - (-3)(-4)]\mathbf{a}_z \\ &= -13\mathbf{a}_x - 14\mathbf{a}_y - 16\mathbf{a}_z \end{aligned}$$